in_emb_y@0_COMP

in_emb_y@0_Y_SEND

in_emb_y@0_X1COMM

in_emb_dw@0_COMP

in_emb_w@1_X2_COMM

in_emb_dw@0_X2_COMM shadow_stack_0_mha_d_x@0_Y_RECV

in_emb_dx@0_COMP

stack_1_mha_q@0_COMP

stack_1_mha_qk@0_COMP

stack_1_mha_d_k@0_COMP

stack_1_mha_q@0_X1COMM

shadow_stack_0_ffn_norm@0_Y_RECV

stack_1_mha_v@0_X1COMM stack_1_mha_k@0_X1COMM

stack_1_mha_res@0_COMP

stack_1_mha_d_wk@0_X2_COMM stack_1_mha_d_wq@0_X2_COMM

stack_1_mha_k@0_COMP stack_1_mha_d_wv@0_X2_COMM

stack_1_mha_d_q@0_COMP

stack_1_mha_v@0_COMP

stack_1_mha_qkv@0_COMP

stack_1_mha_d_qk@0_COMP

stack_1_mha_d_v@0_COMP

stack_1_mha_qkv_reshape@0_COMP

stack_1_mha_norm@0_COMP

stack_1_ffn_x1@0_X1COMM

stack_1_ffn_res@0_COMP

stack_1_ffn_dw1@0_X2_COMM

stack_1_mha_d_res@0_COMP

stack_1_mha_d_qkv@0_COMP

stack_1_mha_d_x34@0_COMP

stack_1_ffn_dx0@0_COMP

stack_1_mha_d_wk@0_COMP stack_1_mha_d_x2@0_COMP stack_1_mha_d_x1@0_COMP stack_1_mha_d_wq@0_COMP stack_1_mha_d_x3@0_COMP stack_1_mha_d_wv@0_COMP

stack_1_mha_wk@1_X2_COMM stack_1_mha_d_x12@0_X2_COMM stack_1_mha_d_x12@0_X1COMM stack_1_mha_wq@1_X2_COMM stack_1_mha_d_x34@0_X1COMM stack_1_mha_wv@1_X2_COMM

stack_1_mha_d_x12@0_COMP

stack_1_mha_d_x@0_COMP

stack_1_ffn_x1@0_COMP stack_1_mha_d_x@0_Y_SEND

stack_1_ffn_dw2@0_COMP stack_1_ffn_x2@0_COMP

stack_1_ffn_res@0_X1COMM

stack_1_ffn_norm@0_COMP

stack_1_ffn_norm@0_Y_SEND

stack_1_ffn_dres@0_COMP

stack_1_ffn_dw2@0_X1COMM

stack_1_ffn_dx1@0_X1COMM

shadow_stack_2_mha_d_x@0_Y_RECV

stack_1_ffn_w2@1_X2_COMM

stack_1_ffn_dx1@0_COMP

stack_1_ffn_dw1@0_COMP

stack_1_ffn_dx01@0_COMP

stack_1_ffn_w1@1_X2_COMM

stack_1_ffn_dx0@0_X2_COMM

stack_3_mha_q@0_COMP

stack_3_mha_qk@0_COMP

stack_3_mha_d_k@0_COMP

stack_3_mha_q@0_X1COMM

shadow_stack_2_ffn_norm@0_Y_RECV

stack_3_mha_v@0_X1COMM stack_3_mha_k@0_X1COMM

stack_3_mha_res@0_COMP

stack_3_mha_d_wk@0_X2_COMM stack_3_mha_d_wq@0_X2_COMM

stack_3_mha_k@0_COMP stack_3_mha_d_wv@0_X2_COMM

stack_3_mha_d_q@0_COMP

stack_3_mha_v@0_COMP

stack_3_mha_qkv@0_COMP

stack_3_mha_d_qk@0_COMP

stack_3_mha_d_v@0_COMP

stack_3_mha_qkv_reshape@0_COMP

stack_3_mha_norm@0_COMP

stack_3_ffn_x1@0_X1COMM

stack_3_ffn_res@0_COMP

stack_3_ffn_dw1@0_X2_COMM

stack_3_mha_d_res@0_COMP

stack_3_mha_d_qkv@0_COMP

stack_3_mha_d_x34@0_COMP

stack_3_ffn_dx0@0_COMP

stack_3_mha_d_wk@0_COMP stack_3_mha_d_x2@0_COMP stack_3_mha_d_x1@0_COMP stack_3_mha_d_wq@0_COMP stack_3_mha_d_x3@0_COMP stack_3_mha_d_wv@0_COMP

stack_3_mha_wk@1_X2_COMM stack_3_mha_d_x12@0_X2_COMM stack_3_mha_d_x12@0_X1COMM stack_3_mha_wq@1_X2_COMM stack_3_mha_d_x34@0_X1COMM stack_3_mha_wv@1_X2_COMM

stack_3_mha_d_x12@0_COMP

stack_3_mha_d_x@0_COMP

stack_3_ffn_x1@0_COMP stack_3_mha_d_x@0_Y_SEND

stack_3_ffn_dw2@0_COMP stack_3_ffn_x2@0_COMP

stack_3_ffn_res@0_X1COMM

stack_3_ffn_norm@0_COMP

stack_3_ffn_norm@0_Y_SEND

stack_3_ffn_dres@0_COMP

stack_3_ffn_dw2@0_X1COMM

stack_3_ffn_dx1@0_X1COMM

shadow_stack_4_mha_d_x@0_Y_RECV

stack_3_ffn_w2@1_X2_COMM

stack_3_ffn_dx1@0_COMP

stack_3_ffn_dw1@0_COMP

stack_3_ffn_dx01@0_COMP

stack_3_ffn_w1@1_X2_COMM

stack_3_ffn_dx0@0_X2_COMM

stack_5_mha_q@0_COMP

stack_5_mha_qk@0_COMP

stack_5_mha_d_k@0_COMP

stack_5_mha_q@0_X1COMM

shadow_stack_4_ffn_norm@0_Y_RECV

stack_5_mha_v@0_X1COMM stack_5_mha_k@0_X1COMM

stack_5_mha_res@0_COMP

stack_5_mha_d_wk@0_X2_COMM stack_5_mha_d_wq@0_X2_COMM

stack_5_mha_k@0_COMP stack_5_mha_d_wv@0_X2_COMM

stack_5_mha_d_q@0_COMP

stack_5_mha_v@0_COMP

stack_5_mha_qkv@0_COMP

stack_5_mha_d_qk@0_COMP

stack_5_mha_d_v@0_COMP

stack_5_mha_qkv_reshape@0_COMP

stack_5_mha_norm@0_COMP

stack_5_ffn_x1@0_X1COMM

stack_5_ffn_res@0_COMP

stack_5_ffn_dw1@0_X2_COMM

stack_5_mha_d_res@0_COMP

stack_5_mha_d_qkv@0_COMP

stack_5_mha_d_x34@0_COMP

stack_5_ffn_dx0@0_COMP

stack_5_mha_d_wk@0_COMP stack_5_mha_d_x2@0_COMP stack_5_mha_d_x1@0_COMP stack_5_mha_d_wq@0_COMP stack_5_mha_d_x3@0_COMP stack_5_mha_d_wv@0_COMP

stack_5_mha_wk@1_X2_COMM stack_5_mha_d_x12@0_X2_COMM stack_5_mha_d_x12@0_X1COMM stack_5_mha_wq@1_X2_COMM stack_5_mha_d_x34@0_X1COMM stack_5_mha_wv@1_X2_COMM

stack_5_mha_d_x12@0_COMP

stack_5_mha_d_x@0_COMP

stack_5_ffn_x1@0_COMP stack_5_mha_d_x@0_Y_SEND

stack_5_ffn_dw2@0_COMP stack_5_ffn_x2@0_COMP

stack_5_ffn_res@0_X1COMM

stack_5_ffn_norm@0_COMP

stack_5_ffn_norm@0_Y_SEND

stack_5_ffn_dres@0_COMP

stack_5_ffn_dw2@0_X1COMM

stack_5_ffn_dx1@0_X1COMM

shadow_stack_6_mha_d_x@0_Y_RECV

stack_5_ffn_w2@1_X2_COMM

stack_5_ffn_dx1@0_COMP

stack_5_ffn_dw1@0_COMP

stack_5_ffn_dx01@0_COMP

stack_5_ffn_w1@1_X2_COMM

stack_5_ffn_dx0@0_X2_COMM

stack_7_mha_q@0_COMP

stack_7_mha_qk@0_COMP

stack_7_mha_d_k@0_COMP

stack_7_mha_q@0_X1COMM

shadow_stack_6_ffn_norm@0_Y_RECV

stack_7_mha_v@0_X1COMM stack_7_mha_k@0_X1COMM

stack_7_mha_res@0_COMP

stack_7_mha_d_wk@0_X2_COMM stack_7_mha_d_wq@0_X2_COMM

stack_7_mha_k@0_COMP stack_7_mha_d_wv@0_X2_COMM

stack_7_mha_d_q@0_COMP

stack_7_mha_v@0_COMP

stack_7_mha_qkv@0_COMP

stack_7_mha_d_qk@0_COMP

stack_7_mha_d_v@0_COMP

stack_7_mha_qkv_reshape@0_COMP

stack_7_mha_norm@0_COMP

stack_7_ffn_x1@0_X1COMM

stack_7_ffn_res@0_COMP

stack_7_ffn_dw1@0_X2_COMM

stack_7_mha_d_res@0_COMP

stack_7_mha_d_qkv@0_COMP

stack_7_mha_d_x34@0_COMP

stack_7_ffn_dx0@0_COMP

stack_7_mha_d_wk@0_COMP stack_7_mha_d_x2@0_COMP stack_7_mha_d_x1@0_COMP stack_7_mha_d_wq@0_COMP stack_7_mha_d_x3@0_COMP stack_7_mha_d_wv@0_COMP

stack_7_mha_wk@1_X2_COMM stack_7_mha_d_x12@0_X2_COMM stack_7_mha_d_x12@0_X1COMM stack_7_mha_wq@1_X2_COMM stack_7_mha_d_x34@0_X1COMM stack_7_mha_wv@1_X2_COMM

stack_7_mha_d_x12@0_COMP

stack_7_mha_d_x@0_COMP

stack_7_ffn_x1@0_COMP stack_7_mha_d_x@0_Y_SEND

stack_7_ffn_dw2@0_COMP stack_7_ffn_x2@0_COMP

stack_7_ffn_res@0_X1COMM

stack_7_ffn_norm@0_COMP

stack_7_ffn_norm@0_Y_SEND

stack_7_ffn_dres@0_COMP

stack_7_ffn_dw2@0_X1COMM

stack_7_ffn_dx1@0_X1COMM

shadow_stack_8_mha_d_x@0_Y_RECV

stack_7_ffn_w2@1_X2_COMM

stack_7_ffn_dx1@0_COMP

stack_7_ffn_dw1@0_COMP

stack_7_ffn_dx01@0_COMP

stack_7_ffn_w1@1_X2_COMM

stack_7_ffn_dx0@0_X2_COMM

stack_9_mha_q@0_COMP

stack_9_mha_qk@0_COMP

stack_9_mha_d_k@0_COMP

stack_9_mha_q@0_X1COMM

shadow_stack_8_ffn_norm@0_Y_RECV

stack_9_mha_v@0_X1COMM stack_9_mha_k@0_X1COMM

stack_9_mha_res@0_COMP

stack_9_mha_d_wk@0_X2_COMM stack_9_mha_d_wq@0_X2_COMM

stack_9_mha_k@0_COMP stack_9_mha_d_wv@0_X2_COMM

stack_9_mha_d_q@0_COMP

stack_9_mha_v@0_COMP

stack_9_mha_qkv@0_COMP

stack_9_mha_d_qk@0_COMP

stack_9_mha_d_v@0_COMP

stack_9_mha_qkv_reshape@0_COMP

stack_9_mha_norm@0_COMP

stack_9_ffn_x1@0_X1COMM

stack_9_ffn_res@0_COMP

stack_9_ffn_dw1@0_X2_COMM

stack_9_mha_d_res@0_COMP

stack_9_mha_d_qkv@0_COMP

stack_9_mha_d_x34@0_COMP

stack_9_ffn_dx0@0_COMP

stack_9_mha_d_wk@0_COMP stack_9_mha_d_x2@0_COMP stack_9_mha_d_x1@0_COMP stack_9_mha_d_wq@0_COMP stack_9_mha_d_x3@0_COMP stack_9_mha_d_wv@0_COMP

stack_9_mha_wk@1_X2_COMM stack_9_mha_d_x12@0_X2_COMM stack_9_mha_d_x12@0_X1COMM stack_9_mha_wq@1_X2_COMM stack_9_mha_d_x34@0_X1COMM stack_9_mha_wv@1_X2_COMM

stack_9_mha_d_x12@0_COMP

stack_9_mha_d_x@0_COMP

stack_9_ffn_x1@0_COMP stack_9_mha_d_x@0_Y_SEND

stack_9_ffn_dw2@0_COMP stack_9_ffn_x2@0_COMP

stack_9_ffn_res@0_X1COMM

stack_9_ffn_norm@0_COMP

stack_9_ffn_norm@0_Y_SEND

stack_9_ffn_dres@0_COMP

stack_9_ffn_dw2@0_X1COMM

stack_9_ffn_dx1@0_X1COMM

shadow_stack_10_mha_d_x@0_Y_RECV

stack_9_ffn_w2@1_X2_COMM

stack_9_ffn_dx1@0_COMP

stack_9_ffn_dw1@0_COMP

stack_9_ffn_dx01@0_COMP

stack_9_ffn_w1@1_X2_COMM

stack_9_ffn_dx0@0_X2_COMM

stack_11_mha_q@0_COMP

stack_11_mha_qk@0_COMP

stack_11_mha_d_k@0_COMP

stack_11_mha_q@0_X1COMM

shadow_stack_10_ffn_norm@0_Y_RECV

stack_11_mha_v@0_X1COMM stack_11_mha_k@0_X1COMM

stack_11_mha_res@0_COMP

stack_11_mha_d_wk@0_X2_COMM stack_11_mha_d_wq@0_X2_COMM

stack_11_mha_k@0_COMP stack_11_mha_d_wv@0_X2_COMM

stack_11_mha_d_q@0_COMP

stack_11_mha_v@0_COMP

stack_11_mha_qkv@0_COMP

stack_11_mha_d_qk@0_COMP

stack_11_mha_d_v@0_COMP

stack_11_mha_qkv_reshape@0_COMP

stack_11_mha_norm@0_COMP

stack_11_ffn_x1@0_X1COMM

stack_11_ffn_res@0_COMP

stack_11_ffn_dw1@0_X2_COMM

stack_11_mha_d_res@0_COMP

stack_11_mha_d_qkv@0_COMP

stack_11_mha_d_x34@0_COMP

stack_11_ffn_dx0@0_COMP

stack_11_mha_d_wk@0_COMP stack_11_mha_d_x2@0_COMP stack_11_mha_d_x1@0_COMP stack_11_mha_d_wq@0_COMP stack_11_mha_d_x3@0_COMP stack_11_mha_d_wv@0_COMP

stack_11_mha_wk@1_X2_COMM stack_11_mha_d_x12@0_X2_COMM stack_11_mha_d_x12@0_X1COMM stack_11_mha_wq@1_X2_COMM stack_11_mha_d_x34@0_X1COMM stack_11_mha_wv@1_X2_COMM

stack_11_mha_d_x12@0_COMP

stack_11_mha_d_x@0_COMP

stack_11_ffn_x1@0_COMP stack_11_mha_d_x@0_Y_SEND

stack_11_ffn_dw2@0_COMP stack_11_ffn_x2@0_COMP

stack_11_ffn_res@0_X1COMM

stack_11_ffn_norm@0_COMP

stack_11_ffn_norm@0_Y_SEND

stack_11_ffn_dres@0_COMP

stack_11_ffn_dw2@0_X1COMM

stack_11_ffn_dx1@0_X1COMM

shadow_stack_12_mha_d_x@0_Y_RECV

stack_11_ffn_w2@1_X2_COMM

stack_11_ffn_dx1@0_COMP

stack_11_ffn_dw1@0_COMP

stack_11_ffn_dx01@0_COMP

stack_11_ffn_w1@1_X2_COMM

stack_11_ffn_dx0@0_X2_COMM

stack_13_mha_q@0_COMP

stack_13_mha_qk@0_COMP

stack_13_mha_d_k@0_COMP

stack_13_mha_q@0_X1COMM

shadow_stack_12_ffn_norm@0_Y_RECV

stack_13_mha_v@0_X1COMM stack_13_mha_k@0_X1COMM

stack_13_mha_res@0_COMP

stack_13_mha_d_wk@0_X2_COMM stack_13_mha_d_wq@0_X2_COMM

stack_13_mha_k@0_COMP stack_13_mha_d_wv@0_X2_COMM

stack_13_mha_d_q@0_COMP

stack_13_mha_v@0_COMP

stack_13_mha_qkv@0_COMP

stack_13_mha_d_qk@0_COMP

stack_13_mha_d_v@0_COMP

stack_13_mha_qkv_reshape@0_COMP

stack_13_mha_norm@0_COMP

stack_13_ffn_x1@0_X1COMM

stack_13_ffn_res@0_COMP

stack_13_ffn_dw1@0_X2_COMM

stack_13_mha_d_res@0_COMP

stack_13_mha_d_qkv@0_COMP

stack_13_mha_d_x34@0_COMP

stack_13_ffn_dx0@0_COMP

stack_13_mha_d_wk@0_COMP stack_13_mha_d_x2@0_COMP stack_13_mha_d_x1@0_COMP stack_13_mha_d_wq@0_COMP stack_13_mha_d_x3@0_COMP stack_13_mha_d_wv@0_COMP

stack_13_mha_wk@1_X2_COMM stack_13_mha_d_x12@0_X2_COMM stack_13_mha_d_x12@0_X1COMM stack_13_mha_wq@1_X2_COMM stack_13_mha_d_x34@0_X1COMM stack_13_mha_wv@1_X2_COMM

stack_13_mha_d_x12@0_COMP

stack_13_mha_d_x@0_COMP

stack_13_ffn_x1@0_COMP stack_13_mha_d_x@0_Y_SEND

stack_13_ffn_dw2@0_COMP stack_13_ffn_x2@0_COMP

stack_13_ffn_res@0_X1COMM

stack_13_ffn_norm@0_COMP

stack_13_ffn_norm@0_Y_SEND

stack_13_ffn_dres@0_COMP

stack_13_ffn_dw2@0_X1COMM

stack_13_ffn_dx1@0_X1COMM

shadow_stack_14_mha_d_x@0_Y_RECV

stack_13_ffn_w2@1_X2_COMM

stack_13_ffn_dx1@0_COMP

stack_13_ffn_dw1@0_COMP

stack_13_ffn_dx01@0_COMP

stack_13_ffn_w1@1_X2_COMM

stack_13_ffn_dx0@0_X2_COMM

stack_15_mha_q@0_COMP

stack_15_mha_qk@0_COMP

stack_15_mha_d_k@0_COMP

stack_15_mha_q@0_X1COMM

shadow_stack_14_ffn_norm@0_Y_RECV

stack_15_mha_v@0_X1COMM stack_15_mha_k@0_X1COMM

stack_15_mha_res@0_COMP

stack_15_mha_d_wk@0_X2_COMM stack_15_mha_d_wq@0_X2_COMM

stack_15_mha_k@0_COMP stack_15_mha_d_wv@0_X2_COMM

stack_15_mha_d_q@0_COMP

stack_15_mha_v@0_COMP

stack_15_mha_qkv@0_COMP

stack_15_mha_d_qk@0_COMP

stack_15_mha_d_v@0_COMP

stack_15_mha_qkv_reshape@0_COMP

stack_15_mha_norm@0_COMP

stack_15_ffn_x1@0_X1COMM

stack_15_ffn_res@0_COMP

stack_15_ffn_dw1@0_X2_COMM

stack_15_mha_d_res@0_COMP

stack_15_mha_d_qkv@0_COMP

stack_15_mha_d_x34@0_COMP

stack_15_ffn_dx0@0_COMP

stack_15_mha_d_wk@0_COMP stack_15_mha_d_x2@0_COMP stack_15_mha_d_x1@0_COMP stack_15_mha_d_wq@0_COMP stack_15_mha_d_x3@0_COMP stack_15_mha_d_wv@0_COMP

stack_15_mha_wk@1_X2_COMM stack_15_mha_d_x12@0_X2_COMM stack_15_mha_d_x12@0_X1COMM stack_15_mha_wq@1_X2_COMM stack_15_mha_d_x34@0_X1COMM stack_15_mha_wv@1_X2_COMM

stack_15_mha_d_x12@0_COMP

stack_15_mha_d_x@0_COMP

stack_15_ffn_x1@0_COMP stack_15_mha_d_x@0_Y_SEND

stack_15_ffn_dw2@0_COMP stack_15_ffn_x2@0_COMP

stack_15_ffn_res@0_X1COMM

stack_15_ffn_norm@0_COMP

stack_15_ffn_norm@0_Y_SEND

stack_15_ffn_dres@0_COMP

stack_15_ffn_dw2@0_X1COMM

stack_15_ffn_dx1@0_X1COMM

shadow_stack_16_mha_d_x@0_Y_RECV

stack_15_ffn_w2@1_X2_COMM

stack_15_ffn_dx1@0_COMP

stack_15_ffn_dw1@0_COMP

stack_15_ffn_dx01@0_COMP

stack_15_ffn_w1@1_X2_COMM

stack_15_ffn_dx0@0_X2_COMM

stack_17_mha_q@0_COMP

stack_17_mha_qk@0_COMP

stack_17_mha_d_k@0_COMP

stack_17_mha_q@0_X1COMM

shadow_stack_16_ffn_norm@0_Y_RECV

stack_17_mha_v@0_X1COMM stack_17_mha_k@0_X1COMM

stack_17_mha_res@0_COMP

stack_17_mha_d_wk@0_X2_COMM stack_17_mha_d_wq@0_X2_COMM

stack_17_mha_k@0_COMP stack_17_mha_d_wv@0_X2_COMM

stack_17_mha_d_q@0_COMP

stack_17_mha_v@0_COMP

stack_17_mha_qkv@0_COMP

stack_17_mha_d_qk@0_COMP

stack_17_mha_d_v@0_COMP

stack_17_mha_qkv_reshape@0_COMP

stack_17_mha_norm@0_COMP

stack_17_ffn_x1@0_X1COMM

stack_17_ffn_res@0_COMP

stack_17_ffn_dw1@0_X2_COMM

stack_17_mha_d_res@0_COMP

stack_17_mha_d_qkv@0_COMP

stack_17_mha_d_x34@0_COMP

stack_17_ffn_dx0@0_COMP

stack_17_mha_d_wk@0_COMP stack_17_mha_d_x2@0_COMP stack_17_mha_d_x1@0_COMP stack_17_mha_d_wq@0_COMP stack_17_mha_d_x3@0_COMP stack_17_mha_d_wv@0_COMP

stack_17_mha_wk@1_X2_COMM stack_17_mha_d_x12@0_X2_COMM stack_17_mha_d_x12@0_X1COMM stack_17_mha_wq@1_X2_COMM stack_17_mha_d_x34@0_X1COMM stack_17_mha_wv@1_X2_COMM

stack_17_mha_d_x12@0_COMP

stack_17_mha_d_x@0_COMP

stack_17_ffn_x1@0_COMP stack_17_mha_d_x@0_Y_SEND

stack_17_ffn_dw2@0_COMP stack_17_ffn_x2@0_COMP

stack_17_ffn_res@0_X1COMM

stack_17_ffn_norm@0_COMP

stack_17_ffn_norm@0_Y_SEND

stack_17_ffn_dres@0_COMP

stack_17_ffn_dw2@0_X1COMM

stack_17_ffn_dx1@0_X1COMM

shadow_stack_18_mha_d_x@0_Y_RECV

stack_17_ffn_w2@1_X2_COMM

stack_17_ffn_dx1@0_COMP

stack_17_ffn_dw1@0_COMP

stack_17_ffn_dx01@0_COMP

stack_17_ffn_w1@1_X2_COMM

stack_17_ffn_dx0@0_X2_COMM

stack_19_mha_q@0_COMP

stack_19_mha_qk@0_COMP

stack_19_mha_d_k@0_COMP

stack_19_mha_q@0_X1COMM

shadow_stack_18_ffn_norm@0_Y_RECV

stack_19_mha_v@0_X1COMM stack_19_mha_k@0_X1COMM

stack_19_mha_res@0_COMP

stack_19_mha_d_wk@0_X2_COMM stack_19_mha_d_wq@0_X2_COMM

stack_19_mha_k@0_COMP stack_19_mha_d_wv@0_X2_COMM

stack_19_mha_d_q@0_COMP

stack_19_mha_v@0_COMP

stack_19_mha_qkv@0_COMP

stack_19_mha_d_qk@0_COMP

stack_19_mha_d_v@0_COMP

stack_19_mha_qkv_reshape@0_COMP

stack_19_mha_norm@0_COMP

stack_19_ffn_x1@0_X1COMM

stack_19_ffn_res@0_COMP

stack_19_ffn_dw1@0_X2_COMM

stack_19_mha_d_res@0_COMP

stack_19_mha_d_qkv@0_COMP

stack_19_mha_d_x34@0_COMP

stack_19_ffn_dx0@0_COMP

stack_19_mha_d_wk@0_COMP stack_19_mha_d_x2@0_COMP stack_19_mha_d_x1@0_COMP stack_19_mha_d_wq@0_COMP stack_19_mha_d_x3@0_COMP stack_19_mha_d_wv@0_COMP

stack_19_mha_wk@1_X2_COMM stack_19_mha_d_x12@0_X2_COMM stack_19_mha_d_x12@0_X1COMM stack_19_mha_wq@1_X2_COMM stack_19_mha_d_x34@0_X1COMM stack_19_mha_wv@1_X2_COMM

stack_19_mha_d_x12@0_COMP

stack_19_mha_d_x@0_COMP

stack_19_ffn_x1@0_COMP stack_19_mha_d_x@0_Y_SEND

stack_19_ffn_dw2@0_COMP stack_19_ffn_x2@0_COMP

stack_19_ffn_res@0_X1COMM

stack_19_ffn_norm@0_COMP

stack_19_ffn_norm@0_Y_SEND

stack_19_ffn_dres@0_COMP

stack_19_ffn_dw2@0_X1COMM

stack_19_ffn_dx1@0_X1COMM

shadow_stack_20_mha_d_x@0_Y_RECV

stack_19_ffn_w2@1_X2_COMM

stack_19_ffn_dx1@0_COMP

stack_19_ffn_dw1@0_COMP

stack_19_ffn_dx01@0_COMP

stack_19_ffn_w1@1_X2_COMM

stack_19_ffn_dx0@0_X2_COMM

stack_21_mha_q@0_COMP

stack_21_mha_qk@0_COMP

stack_21_mha_d_k@0_COMP

stack_21_mha_q@0_X1COMM

shadow_stack_20_ffn_norm@0_Y_RECV

stack_21_mha_v@0_X1COMM stack_21_mha_k@0_X1COMM

stack_21_mha_res@0_COMP

stack_21_mha_d_wk@0_X2_COMM stack_21_mha_d_wq@0_X2_COMM

stack_21_mha_k@0_COMP stack_21_mha_d_wv@0_X2_COMM

stack_21_mha_d_q@0_COMP

stack_21_mha_v@0_COMP

stack_21_mha_qkv@0_COMP

stack_21_mha_d_qk@0_COMP

stack_21_mha_d_v@0_COMP

stack_21_mha_qkv_reshape@0_COMP

stack_21_mha_norm@0_COMP

stack_21_ffn_x1@0_X1COMM

stack_21_ffn_res@0_COMP

stack_21_ffn_dw1@0_X2_COMM

stack_21_mha_d_res@0_COMP

stack_21_mha_d_qkv@0_COMP

stack_21_mha_d_x34@0_COMP

stack_21_ffn_dx0@0_COMP

stack_21_mha_d_wk@0_COMP stack_21_mha_d_x2@0_COMP stack_21_mha_d_x1@0_COMP stack_21_mha_d_wq@0_COMP stack_21_mha_d_x3@0_COMP stack_21_mha_d_wv@0_COMP

stack_21_mha_wk@1_X2_COMM stack_21_mha_d_x12@0_X2_COMM stack_21_mha_d_x12@0_X1COMM stack_21_mha_wq@1_X2_COMM stack_21_mha_d_x34@0_X1COMM stack_21_mha_wv@1_X2_COMM

stack_21_mha_d_x12@0_COMP

stack_21_mha_d_x@0_COMP

stack_21_ffn_x1@0_COMP stack_21_mha_d_x@0_Y_SEND

stack_21_ffn_dw2@0_COMP stack_21_ffn_x2@0_COMP

stack_21_ffn_res@0_X1COMM

stack_21_ffn_norm@0_COMP

stack_21_ffn_norm@0_Y_SEND

stack_21_ffn_dres@0_COMP

stack_21_ffn_dw2@0_X1COMM

stack_21_ffn_dx1@0_X1COMM

shadow_stack_22_mha_d_x@0_Y_RECV

stack_21_ffn_w2@1_X2_COMM

stack_21_ffn_dx1@0_COMP

stack_21_ffn_dw1@0_COMP

stack_21_ffn_dx01@0_COMP

stack_21_ffn_w1@1_X2_COMM

stack_21_ffn_dx0@0_X2_COMM

stack_23_mha_q@0_COMP

stack_23_mha_qk@0_COMP

stack_23_mha_d_k@0_COMP

stack_23_mha_q@0_X1COMM

shadow_stack_22_ffn_norm@0_Y_RECV

stack_23_mha_v@0_X1COMM stack_23_mha_k@0_X1COMM

stack_23_mha_res@0_COMP

stack_23_mha_d_wk@0_X2_COMM stack_23_mha_d_wq@0_X2_COMM

stack_23_mha_k@0_COMP stack_23_mha_d_wv@0_X2_COMM

stack_23_mha_d_q@0_COMP

stack_23_mha_v@0_COMP

stack_23_mha_qkv@0_COMP

stack_23_mha_d_qk@0_COMP

stack_23_mha_d_v@0_COMP

stack_23_mha_qkv_reshape@0_COMP

stack_23_mha_norm@0_COMP

stack_23_ffn_x1@0_X1COMM

stack_23_ffn_res@0_COMP

stack_23_ffn_dw1@0_X2_COMM

stack_23_mha_d_res@0_COMP

stack_23_mha_d_qkv@0_COMP

stack_23_mha_d_x34@0_COMP

stack_23_ffn_dx0@0_COMP

stack_23_mha_d_wk@0_COMP stack_23_mha_d_x2@0_COMP stack_23_mha_d_x1@0_COMP stack_23_mha_d_wq@0_COMP stack_23_mha_d_x3@0_COMP stack_23_mha_d_wv@0_COMP

stack_23_mha_wk@1_X2_COMM stack_23_mha_d_x12@0_X2_COMM stack_23_mha_d_x12@0_X1COMM stack_23_mha_wq@1_X2_COMM stack_23_mha_d_x34@0_X1COMM stack_23_mha_wv@1_X2_COMM

stack_23_mha_d_x12@0_COMP

stack_23_mha_d_x@0_COMP

stack_23_ffn_x1@0_COMP stack_23_mha_d_x@0_Y_SEND

stack_23_ffn_dw2@0_COMP stack_23_ffn_x2@0_COMP

stack_23_ffn_res@0_X1COMM

stack_23_ffn_norm@0_COMP

stack_23_ffn_norm@0_Y_SEND

stack_23_ffn_dres@0_COMP

stack_23_ffn_dw2@0_X1COMM

stack_23_ffn_dx1@0_X1COMM

shadow_stack_24_mha_d_x@0_Y_RECV

stack_23_ffn_w2@1_X2_COMM

stack_23_ffn_dx1@0_COMP

stack_23_ffn_dw1@0_COMP

stack_23_ffn_dx01@0_COMP

stack_23_ffn_w1@1_X2_COMM

stack_23_ffn_dx0@0_X2_COMM

stack_25_mha_q@0_COMP

stack_25_mha_qk@0_COMP

stack_25_mha_d_k@0_COMP

stack_25_mha_q@0_X1COMM

shadow_stack_24_ffn_norm@0_Y_RECV

stack_25_mha_v@0_X1COMM stack_25_mha_k@0_X1COMM

stack_25_mha_res@0_COMP

stack_25_mha_d_wk@0_X2_COMM stack_25_mha_d_wq@0_X2_COMM

stack_25_mha_k@0_COMP stack_25_mha_d_wv@0_X2_COMM

stack_25_mha_d_q@0_COMP

stack_25_mha_v@0_COMP

stack_25_mha_qkv@0_COMP

stack_25_mha_d_qk@0_COMP

stack_25_mha_d_v@0_COMP

stack_25_mha_qkv_reshape@0_COMP

stack_25_mha_norm@0_COMP

stack_25_ffn_x1@0_X1COMM

stack_25_ffn_res@0_COMP

stack_25_ffn_dw1@0_X2_COMM

stack_25_mha_d_res@0_COMP

stack_25_mha_d_qkv@0_COMP

stack_25_mha_d_x34@0_COMP

stack_25_ffn_dx0@0_COMP

stack_25_mha_d_wk@0_COMP stack_25_mha_d_x2@0_COMP stack_25_mha_d_x1@0_COMP stack_25_mha_d_wq@0_COMP stack_25_mha_d_x3@0_COMP stack_25_mha_d_wv@0_COMP

stack_25_mha_wk@1_X2_COMM stack_25_mha_d_x12@0_X2_COMM stack_25_mha_d_x12@0_X1COMM stack_25_mha_wq@1_X2_COMM stack_25_mha_d_x34@0_X1COMM stack_25_mha_wv@1_X2_COMM

stack_25_mha_d_x12@0_COMP

stack_25_mha_d_x@0_COMP

stack_25_ffn_x1@0_COMP stack_25_mha_d_x@0_Y_SEND

stack_25_ffn_dw2@0_COMP stack_25_ffn_x2@0_COMP

stack_25_ffn_res@0_X1COMM

stack_25_ffn_norm@0_COMP

stack_25_ffn_norm@0_Y_SEND

stack_25_ffn_dres@0_COMP

stack_25_ffn_dw2@0_X1COMM

stack_25_ffn_dx1@0_X1COMM

shadow_stack_26_mha_d_x@0_Y_RECV

stack_25_ffn_w2@1_X2_COMM

stack_25_ffn_dx1@0_COMP

stack_25_ffn_dw1@0_COMP

stack_25_ffn_dx01@0_COMP

stack_25_ffn_w1@1_X2_COMM

stack_25_ffn_dx0@0_X2_COMM

stack_27_mha_q@0_COMP

stack_27_mha_qk@0_COMP

stack_27_mha_d_k@0_COMP

stack_27_mha_q@0_X1COMM

shadow_stack_26_ffn_norm@0_Y_RECV

stack_27_mha_v@0_X1COMM stack_27_mha_k@0_X1COMM

stack_27_mha_res@0_COMP

stack_27_mha_d_wk@0_X2_COMM stack_27_mha_d_wq@0_X2_COMM

stack_27_mha_k@0_COMP stack_27_mha_d_wv@0_X2_COMM

stack_27_mha_d_q@0_COMP

stack_27_mha_v@0_COMP

stack_27_mha_qkv@0_COMP

stack_27_mha_d_qk@0_COMP

stack_27_mha_d_v@0_COMP

stack_27_mha_qkv_reshape@0_COMP

stack_27_mha_norm@0_COMP

stack_27_ffn_x1@0_X1COMM

stack_27_ffn_res@0_COMP

stack_27_ffn_dw1@0_X2_COMM

stack_27_mha_d_res@0_COMP

stack_27_mha_d_qkv@0_COMP

stack_27_mha_d_x34@0_COMP

stack_27_ffn_dx0@0_COMP

stack_27_mha_d_wk@0_COMP stack_27_mha_d_x2@0_COMP stack_27_mha_d_x1@0_COMP stack_27_mha_d_wq@0_COMP stack_27_mha_d_x3@0_COMP stack_27_mha_d_wv@0_COMP

stack_27_mha_wk@1_X2_COMM stack_27_mha_d_x12@0_X2_COMM stack_27_mha_d_x12@0_X1COMM stack_27_mha_wq@1_X2_COMM stack_27_mha_d_x34@0_X1COMM stack_27_mha_wv@1_X2_COMM

stack_27_mha_d_x12@0_COMP

stack_27_mha_d_x@0_COMP

stack_27_ffn_x1@0_COMP stack_27_mha_d_x@0_Y_SEND

stack_27_ffn_dw2@0_COMP stack_27_ffn_x2@0_COMP

stack_27_ffn_res@0_X1COMM

stack_27_ffn_norm@0_COMP

stack_27_ffn_norm@0_Y_SEND

stack_27_ffn_dres@0_COMP

stack_27_ffn_dw2@0_X1COMM

stack_27_ffn_dx1@0_X1COMM

shadow_stack_28_mha_d_x@0_Y_RECV

stack_27_ffn_w2@1_X2_COMM

stack_27_ffn_dx1@0_COMP

stack_27_ffn_dw1@0_COMP

stack_27_ffn_dx01@0_COMP

stack_27_ffn_w1@1_X2_COMM

stack_27_ffn_dx0@0_X2_COMM

stack_29_mha_q@0_COMP

stack_29_mha_qk@0_COMP

stack_29_mha_d_k@0_COMP

stack_29_mha_q@0_X1COMM

shadow_stack_28_ffn_norm@0_Y_RECV

stack_29_mha_v@0_X1COMM stack_29_mha_k@0_X1COMM

stack_29_mha_res@0_COMP

stack_29_mha_d_wk@0_X2_COMM stack_29_mha_d_wq@0_X2_COMM

stack_29_mha_k@0_COMP stack_29_mha_d_wv@0_X2_COMM

stack_29_mha_d_q@0_COMP

stack_29_mha_v@0_COMP

stack_29_mha_qkv@0_COMP

stack_29_mha_d_qk@0_COMP

stack_29_mha_d_v@0_COMP

stack_29_mha_qkv_reshape@0_COMP

stack_29_mha_norm@0_COMP

stack_29_ffn_x1@0_X1COMM

stack_29_ffn_res@0_COMP

stack_29_ffn_dw1@0_X2_COMM

stack_29_mha_d_res@0_COMP

stack_29_mha_d_qkv@0_COMP

stack_29_mha_d_x34@0_COMP

stack_29_ffn_dx0@0_COMP

stack_29_mha_d_wk@0_COMP stack_29_mha_d_x2@0_COMP stack_29_mha_d_x1@0_COMP stack_29_mha_d_wq@0_COMP stack_29_mha_d_x3@0_COMP stack_29_mha_d_wv@0_COMP

stack_29_mha_wk@1_X2_COMM stack_29_mha_d_x12@0_X2_COMM stack_29_mha_d_x12@0_X1COMM stack_29_mha_wq@1_X2_COMM stack_29_mha_d_x34@0_X1COMM stack_29_mha_wv@1_X2_COMM

stack_29_mha_d_x12@0_COMP

stack_29_mha_d_x@0_COMP

stack_29_ffn_x1@0_COMP stack_29_mha_d_x@0_Y_SEND

stack_29_ffn_dw2@0_COMP stack_29_ffn_x2@0_COMP

stack_29_ffn_res@0_X1COMM

stack_29_ffn_norm@0_COMP

stack_29_ffn_norm@0_Y_SEND

stack_29_ffn_dres@0_COMP

stack_29_ffn_dw2@0_X1COMM

stack_29_ffn_dx1@0_X1COMM

shadow_stack_30_mha_d_x@0_Y_RECV

stack_29_ffn_w2@1_X2_COMM

stack_29_ffn_dx1@0_COMP

stack_29_ffn_dw1@0_COMP

stack_29_ffn_dx01@0_COMP

stack_29_ffn_w1@1_X2_COMM

stack_29_ffn_dx0@0_X2_COMM

stack_31_mha_q@0_COMP

stack_31_mha_qk@0_COMP

stack_31_mha_d_k@0_COMP

stack_31_mha_q@0_X1COMM

shadow_stack_30_ffn_norm@0_Y_RECV

stack_31_mha_v@0_X1COMM stack_31_mha_k@0_X1COMM

stack_31_mha_res@0_COMP

stack_31_mha_d_wk@0_X2_COMM stack_31_mha_d_wq@0_X2_COMM

stack_31_mha_k@0_COMP stack_31_mha_d_wv@0_X2_COMM

stack_31_mha_d_q@0_COMP

stack_31_mha_v@0_COMP

stack_31_mha_qkv@0_COMP

stack_31_mha_d_qk@0_COMP

stack_31_mha_d_v@0_COMP

stack_31_mha_qkv_reshape@0_COMP

stack_31_mha_norm@0_COMP

stack_31_ffn_x1@0_X1COMM

stack_31_ffn_res@0_COMP

stack_31_ffn_dw1@0_X2_COMM

stack_31_mha_d_res@0_COMP

stack_31_mha_d_qkv@0_COMP

stack_31_mha_d_x34@0_COMP

stack_31_ffn_dx0@0_COMP

stack_31_mha_d_wk@0_COMP stack_31_mha_d_x2@0_COMP stack_31_mha_d_x1@0_COMP stack_31_mha_d_wq@0_COMP stack_31_mha_d_x3@0_COMP stack_31_mha_d_wv@0_COMP

stack_31_mha_wk@1_X2_COMM stack_31_mha_d_x12@0_X2_COMM stack_31_mha_d_x12@0_X1COMM stack_31_mha_wq@1_X2_COMM stack_31_mha_d_x34@0_X1COMM stack_31_mha_wv@1_X2_COMM

stack_31_mha_d_x12@0_COMP

stack_31_mha_d_x@0_COMP

stack_31_ffn_x1@0_COMP stack_31_mha_d_x@0_Y_SEND

stack_31_ffn_dw2@0_COMP stack_31_ffn_x2@0_COMP

stack_31_ffn_res@0_X1COMM

stack_31_ffn_norm@0_COMP

stack_31_ffn_norm@0_Y_SEND

stack_31_ffn_dnorm@0_X1COMM

stack_31_ffn_dres@0_COMP

shadow_out_emb_dx@0_Y_RECV

stack_31_ffn_dw2@0_X1COMM

stack_31_ffn_dx1@0_X1COMM

stack_31_ffn_w2@1_X2_COMM

stack_31_ffn_dx1@0_COMP

stack_31_ffn_dw1@0_COMP

stack_31_ffn_dx01@0_COMP

stack_31_ffn_w1@1_X2_COMM

stack_31_ffn_dx0@0_X2_COMM

in_emb_w@1_COMP

stack_1_mha_wk@1_COMP stack_1_mha_wq@1_COMP stack_1_mha_wv@1_COMP

stack_1_ffn_w1@1_COMP

stack_1_ffn_w2@1_COMP

stack_3_mha_wk@1_COMP stack_3_mha_wq@1_COMP stack_3_mha_wv@1_COMP

stack_3_ffn_w1@1_COMP

stack_3_ffn_w2@1_COMP

stack_5_mha_wk@1_COMP stack_5_mha_wq@1_COMP stack_5_mha_wv@1_COMP

stack_5_ffn_w1@1_COMP

stack_5_ffn_w2@1_COMP

stack_7_mha_wk@1_COMP stack_7_mha_wq@1_COMP stack_7_mha_wv@1_COMP

stack_7_ffn_w1@1_COMP

stack_7_ffn_w2@1_COMP

stack_9_mha_wk@1_COMP stack_9_mha_wq@1_COMP stack_9_mha_wv@1_COMP

stack_9_ffn_w1@1_COMP

stack_9_ffn_w2@1_COMP

stack_11_mha_wk@1_COMP stack_11_mha_wq@1_COMP stack_11_mha_wv@1_COMP

stack_11_ffn_w1@1_COMP

stack_11_ffn_w2@1_COMP

stack_13_mha_wk@1_COMP stack_13_mha_wq@1_COMP stack_13_mha_wv@1_COMP

stack_13_ffn_w1@1_COMP

stack_13_ffn_w2@1_COMP

stack_15_mha_wk@1_COMP stack_15_mha_wq@1_COMP stack_15_mha_wv@1_COMP

stack_15_ffn_w1@1_COMP

stack_15_ffn_w2@1_COMP

stack_17_mha_wk@1_COMP stack_17_mha_wq@1_COMP stack_17_mha_wv@1_COMP

stack_17_ffn_w1@1_COMP

stack_17_ffn_w2@1_COMP

stack_19_mha_wk@1_COMP stack_19_mha_wq@1_COMP stack_19_mha_wv@1_COMP

stack_19_ffn_w1@1_COMP

stack_19_ffn_w2@1_COMP

stack_21_mha_wk@1_COMP stack_21_mha_wq@1_COMP stack_21_mha_wv@1_COMP

stack_21_ffn_w1@1_COMP

stack_21_ffn_w2@1_COMP

stack_23_mha_wk@1_COMP stack_23_mha_wq@1_COMP stack_23_mha_wv@1_COMP

stack_23_ffn_w1@1_COMP

stack_23_ffn_w2@1_COMP

stack_25_mha_wk@1_COMP stack_25_mha_wq@1_COMP stack_25_mha_wv@1_COMP

stack_25_ffn_w1@1_COMP

stack_25_ffn_w2@1_COMP

stack_27_mha_wk@1_COMP stack_27_mha_wq@1_COMP stack_27_mha_wv@1_COMP

stack_27_ffn_w1@1_COMP

stack_27_ffn_w2@1_COMP

stack_29_mha_wk@1_COMP stack_29_mha_wq@1_COMP stack_29_mha_wv@1_COMP

stack_29_ffn_w1@1_COMP

stack_29_ffn_w2@1_COMP

stack_31_mha_wk@1_COMP stack_31_mha_wq@1_COMP stack_31_mha_wv@1_COMP

stack_31_ffn_w1@1_COMP

stack_31_ffn_w2@1_COMP